UE03 Pokemon ER Diagram

Hypermedia
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Practical Exercise

In this assignment we attempt create an Entity Relationship model:

- Construct an ER diagram using the **Chen Notation**
- List any assumtions you make while modeling
- You can use modelling tools such as `https://github.com/borkdominik/bigER` or `https://dreampuf.github.io/GraphvizOnline/` or pen and paper.

Exercise 1 *The Pokémon World League is building a database to keep track of its registered Pokémon and Trainers.*

Each Pokémon has a unique number in the Pokédex, a name, and the generation it first appeared in. Some Pokémon can evolve from other Pokémon through specific methods such as leveling up, using an item, or meeting a special condition.

Pokémon can have one or more special abilities that influence their behavior in battles or in the world. Additionally, each Pokémon has one or two elemental types (such as Fire, Water, Grass) that determine its strengths and weaknesses.

Some types are particularly effective against others; for example, Water attacks are effective against Fire types. The system must record which types are strong against which other types.

Trainers collect Pokémon and organize them into Teams. Each Team has a name and can consist of 1 to 6 Pokémon. Within a Team, each Pokémon occupies a specific position (first, second, etc.) and has a recorded level. Additionally, Trainers can assign a held item to a Pokémon for use in battles while the Pokémon is part of a Team.

Trainers participate in regional leagues and can earn special badges by defeating Gym Leaders.

Exercise 2 (Advanced) *The World League is planning additional features for its system and needs the design to be extended accordingly.*

Pokémon can learn a variety of moves. Moves have names, belong to a particular elemental type, and have a certain amount of power. A Pokémon may learn many different moves during its lifetime.

Trainers can own items to assist them, such as healing potions, Poké Balls, or evolutionary stones.

The League also wants to record battles between Trainers. For each battle, the system should capture the date, the names of the two Trainers involved, and which Trainer won. Optionally, it should also keep track of which Pokémon participated in each battle.

Each Pokémon has various inherent attributes: health points, attack strength, defense capability, special attack and special defense statistics, and speed. These attributes have a base for a species but individual pokemon may vary.

Additionally, the League recognizes that some Pokémon exist in regional forms, such as Alolan or Galarian variants. A Pokémon may also develop a closer bond with its Trainer over time, measured by a friendship level, which can affect its evolution possibilities.

Hint Sheet: Pokémon Database Assignment

Need some help? Try thinking about these questions:

Understanding the World

- What are the main *things* (objects or people) the system is keeping track of?
- Are there things that *belong* to other things (e.g., a Pokémon belonging to a Trainer)?
- When the text mentions ”can have” or ”may belong to”, what does that suggest about relationships and cardinalities?

Finding Entities

- Which nouns in the description seem like they represent important *things*?
- Is something unique or identifiable? (If yes, it's probably an entity.)

Finding Relationships

- When two things are mentioned together (e.g., Pokémon and Teams), is there a *connection* or *ownership*?
- Are there numbers or quantities mentioned (like ”1 to 6”)? What does that tell you about how many items are connected?

Attributes vs. Entities

- Is something describing a property (like ”Name” or ”Level”)?
  ⇒ It's probably an attribute.
- Is something a full ”object” you could describe separately?
  ⇒ It's probably an entity.

Special Structures

- If something changes over time (like battles or evolution), does it need its own entity or relationship?
- Are there optional features? (e.g., not every Pokémon evolves.)
- Are there multiple levels of relationships (e.g., Teams owned by Trainers, Pokémon inside Teams)?

Bonus (Advanced Extension)

- Moves: How do Pokémon and Moves relate?
- Items: Who owns them? Who holds them?
- Battles: What needs to be recorded beyond just the participants?

Final Advice:

- Start simple: identify a few key entities first.
- Draw rough relationship lines between them.
- Only then think about attributes.
- Cardinalities (one-to-many, many-to-many) are important — pay attention to words like ”can have multiple”, ”only one”, or ”optional”.